Physics
Standard and higher level
Paper 1A and Paper 2
B.3 Gas laws

12 October 2026

W4ss Academy practice test 3 | **Topic** B.3 Gas laws | **Ref.** W4SS–PHY–B3–03

Student name

30 minutes [Paper 1A and Paper 2]

Instructions to students

- Write your name in the boxes above.
- Do not open this examination paper until instructed to do so.
- Answer all questions.
- **Paper 1A:** for each question, choose the answer you consider to be the best and indicate your choice on the answer sheet provided.
- **Paper 2:** answers must be written within the answer boxes provided.
- A calculator is required for this paper.
- A clean copy of the **physics data booklet** is required for this paper.
- The maximum mark for paper 1A is **[10 marks]**.
- The maximum mark for paper 2 is **[10 marks]**.
- The maximum mark for this examination paper is **[20 marks]**.

Paper 1A

1. The equation $\frac{pV}{T}$ = constant is applied to a real gas where p is the pressure of the gas, V is its volume and T is its temperature.

What is correct about this equation?

A. It is empirical.

B. It is theoretical.

C. It cannot be tested.

D. It cannot be disproved.

2. What is **not** an assumption of the kinetic model of an ideal gas?

A. Attractive forces between molecules are negligible.

B. Collision duration is negligible compared with time between collisions.

C. Molecules suffer negligible momentum change during wall collisions.

D. Molecular volume is negligible compared with gas volume.

3. Which aspect of thermal physics is best explained by the molecular kinetic model?

A. The equation of state of ideal gases

B. The difference between Celsius and Kelvin temperature

C. The value of the Avogadro constant

D. The existence of gaseous isotopes

4. Two samples of a gas are kept in separate containers. The molecules of each sample have the same average translational speed, but the samples have a different density.

What is correct about the pressure and the temperature of the samples, as compared to each other?

	Pressure	Temperature
A.	same	same
B.	same	different
C.	different	same
D.	different	different

5. A sample of a gas has volume V and contains N molecules, each of mass m. The average translational speed of the molecules is u.

Which expression is equivalent to the pressure of the gas?

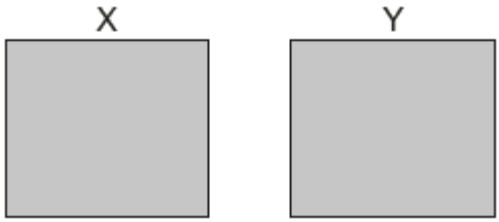
A. $\frac{1}{3}\frac{NV}{m}u^2$

B. $\frac{1}{3}\frac{Nm}{V}u^2$

C. $\frac{1}{3}\frac{N}{mV}u^2$

D. $\frac{1}{3}\frac{m}{NV}u^2$

6. Two identical containers X and Y each contain an ideal gas. X has N molecules of gas at an absolute temperature of T and Y has $3N$ molecules of gas at an absolute temperature of $\frac{T}{2}$. What is the ratio of the pressures $\frac{P_Y}{P_X}$?



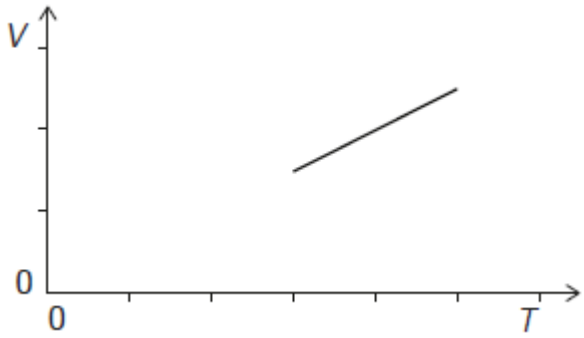
A. $\frac{1}{6}$

B. $\frac{2}{3}$

C. $\frac{3}{2}$

D. 6

7. The molar mass of an ideal gas is M. A fixed mass m of the gas expands at a constant pressure p. The graph shows the variation with temperature T of the gas volume V.



What is the gradient of the graph?

A. $\frac{Mp}{mR}$

B. $\frac{MR}{mp}$

C. $\frac{mp}{MR}$

D. $\frac{mR}{Mp}$

8. Two containers, X and Y, are filled with an ideal gas at the same pressure.

The volume of X is four times the volume of Y. The temperature of X is 327 °C and the temperature of Y is 27 °C.

What is $\frac{\text{amount of substance in X}}{\text{amount of substance in Y}}$?

A. $\frac{1}{8}$

B. $\frac{1}{2}$

C. 2

D. 8

9. Cylinder X has a volume V and contains 3.0 mol of an ideal gas. Cylinder Y has a volume $\frac{V}{2}$ and contains 2.0 mol of the same gas.

The gases in X and Y are at the same temperature T. The containers are joined by a valve which is opened so that the temperatures do not change.

What is the change in pressure in X?

A. $+\frac{1}{3}(\frac{RT}{V})$

B. $-\frac{1}{3}(\frac{RT}{V})$

C. $+\frac{2}{3}(\frac{RT}{V})$

D. $-\frac{2}{3}(\frac{RT}{V})$

10. The density of an ideal gas is 1.4 kg m^{-3} when its pressure is 0.1 MPa.

What is the average translational speed of the gas molecules?

A. 0.46 m s^{-1}

B. 270 m s^{-1}

C. 460 m s^{-1}

D. 71 km s^{-1}

Paper 1A — answer sheet

Student name _____ Date _____

1.	A	B	C	D
2.	A	B	C	D
3.	A	B	C	D
4.	A	B	C	D
5.	A	B	C	D
6.	A	B	C	D
7.	A	B	C	D
8.	A	B	C	D
9.	A	B	C	D
10.	A	B	C	D

Shade **one** circle per question. If you change your answer, cross out the wrong circle clearly.

Paper 2

Answer **all** questions. Answers must be written within the answer boxes provided.

1. (a) State **two** assumptions of the kinetic model of an ideal gas that refer to intermolecular collisions. [2]

(b) Discuss how the motion of the molecules of a gas gives rise to pressure in the gas. [3]

2. (a) Outline how the concept of absolute zero of temperature is interpreted in terms of:

(i) the ideal gas law, [1]

(ii) the kinetic energy of particles in an ideal gas. [1]

(This question continues on the following page)

(Question 2 continued)

A container holds a mixture of argon and helium atoms at a temperature of 37 °C.

(b) Discuss how the mean kinetic energy of the argon atoms in the mixture compares with that of the helium atoms. [2]

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3. State **one** way in which a real gas differs from an ideal gas. [1]

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Sources of the questions

This practice test was put together by W4ss Academy for internal class use. The questions are taken from the IB DP Physics Questionbank (first assessment 2025), topic B.3 Gas laws. They remain the property of the International Baccalaureate Organization and must not be published or sold.

Paper 1A — question 1	19M.1A.SL.TZ2.13
Paper 1A — question 2	20N.1A.SL.TZ0.7
Paper 1A — question 3	21M.1A.SL.TZ1.10 (variant 1)
Paper 1A — question 4	EXE.1A.SL.TZ0.9
Paper 1A — question 5	EXE.1A.SL.TZ0.10
Paper 1A — question 6	21M.1A.SL.TZ2.11
Paper 1A — question 7	21N.1A.SL.TZ0.10
Paper 1A — question 8	SPM.1A.HL.TZ0.17
Paper 1A — question 9	19M.1A.SL.TZ2.14
Paper 1A — question 10	EXE.1A.SL.TZ0.12
Paper 2 — question 1	EXE.2.SL.TZ0.7 (parts (a), (b))
Paper 2 — question 2	EXE.2.SL.TZ0.9 (parts (a)(i), (a)(ii), (c))
Paper 2 — question 3	22N.2.SL.TZ0.2 (part (b)(i))

Please **do not** write on this page.

Answers written on this page
will not be marked.